AMS526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 18: Sensitivity of Eigenvalues; Review for Midterm #2

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Outline

Sensitivity of Eigenvalues

- Condition number of matrix X determines sensitivity of eigenvalues

Theorem

Let $A \in \mathbb{C}^{n\times n}$ be a nondefective matrix, and suppose $A = X\Lambda X^{-1}$, where X is nonsingular and Λ is diagonal. Let $\delta A \in \mathbb{C}^{n\times n}$ be some perturbation of A, and let μ be an eigenvalue of $A + \delta A$. Then A has an eigenvalue λ such that

$$|\mu - \lambda| \leq \kappa_p(X)\|\delta A\|_p$$

for $1 \leq p \leq \infty$.

- $\kappa_p(X)$ measures how far eigenvectors are from linear dependence
- For normal matrices, condition number $\kappa_2(X) = 1$ and $\kappa_p(X) = O(1)$, so eigenvalues of normal matrices are always well-conditioned

Sensitivity of Eigenvalues

Proof.

Let $\delta\Lambda = X^{-1}(\delta A)X$. Then

$$\|\delta\Lambda\|_p \leq \|X^{-1}\|_p \|\delta A\|_p \|X\|_p = \kappa_p(X)\|\delta A\|_p.$$

Let y be an engenvector of $\Lambda + \delta\Lambda$ associated with μ. Suppose μ is not an eigenvalue of A, so $\mu I - \Lambda$ is nonsingular.
$(\Lambda + \delta\Lambda)y = \mu y \Rightarrow (\mu I - \Lambda)y = (\delta\Lambda)\,y \Rightarrow y = (\mu I - \Lambda)^{-1}(\delta\Lambda)\,y$. Thus

$$\|(\mu I - \Lambda)^{-1}\|_p^{-1} \leq \|\delta\Lambda\|_p.$$

$\|(\mu I - \Lambda)^{-1}\|_p = |\mu - \lambda|^{-1}$, where λ is the eigenvalue of A closest to μ. Thus,

$$|\mu - \lambda| \leq \|\delta\Lambda\|_p \leq \kappa_p(X)\|\delta A\|_p.$$

□

Left and Right Eigenvectors

- To analyze sensitivity of individual eigenvalues, we need to define left and right eigenvectors
  - $Ax = \lambda x$ for nonzero x then x is *right eigenvector* associated with λ
  - $y^* A = \lambda y^*$ for nonzero y, then y is *left eigenvector* associated with λ
- Left eigenvectors of A are right eigenvectors of A^*

Theorem

Let $A \in \mathbb{C}^{n\times n}$ have distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ with associated linearly independent right eigenvectors $x_1, \ldots, x_n$ and left eigenvectors $y_1, \ldots, y_n$. Then $y_j^* x_i \neq 0$ if $i = j$ and $y_j^* x_i = 0$ if $i \neq j$.

Proof.

If $i \neq j$, $y_j^* A x_i = \lambda_i y_j^* x_i$ and $y_j^* A x_i = \lambda_j y_j^* x_i$. Since $\lambda_i \neq \lambda_j$, $y_j^* x_i = 0$. If $i = j$, since $\{x_i\}$ form a basis for $\mathbb{C}^n$, $y_i^* x_i = 0$ together with $y_i^* x_j = 0$ would imply that $y_i = 0$. This leads to a contradiction. $\square$

Sensitivity of Individual Eigenvalues

- We analyze sensitivity of individual eigenvalues that are distinct

Theorem

Let $A \in \mathbb{C}^{n\times n}$ have n distinct eigenvalues. Let λ be an eigenvalue with associated right and left eigenvectors x and y, respectively, normalized so that $\|x\|_2 = \|y\|_2 = 1$. Let δA be a small perturbation satisfying $\|\delta A\|_2 = \epsilon$, and let $\lambda + \delta\lambda$ be the eigenvalue of $A + \delta A$ that approximates λ. Then

$$|\delta\lambda| \leq \frac{1}{|y^* x|}\epsilon + O(\epsilon^2).$$

- $\kappa = 1/|y^* x|$ is condition number for eigenvalue λ
- A simple eigenvalue is sensitive if its associated right and left eigenvectors are nearly orthogonal

Sensitivity of Individual Eigenvalues

Proof.

We know that $|\delta\lambda| \leq \kappa_p(X)\epsilon = O(\epsilon)$. In addition, $\delta x = O(\epsilon)$ when λ is a simple eigenvalue (proof omitted). Because $(A + \delta A)(x + \delta x) = (\lambda + \delta\lambda)(x + \delta x)$, thus

$$(\delta A)\, x + A\,(\delta x) + O(\epsilon^2) = (\delta\lambda)x + \lambda(\delta x) + O(\epsilon^2).$$

Left multiplying by y^* and using equation $y^*A = \lambda y^*$, we obtain

$$y^*\,(\delta A)\, x + O(\epsilon^2) = (\delta\lambda)\, y^* x + O(\epsilon^2)$$

and hence

$$\delta\lambda = \frac{y^*\,(\delta A)\, x}{y^* x} + O(\epsilon^2).$$

Since $|y^*\,(\delta A)\, x| \leq \|y\|_2\, \|(\delta A)\|_2\, \|x\|_2 = \epsilon$, $|\delta\lambda| \leq \frac{1}{|y^* x|}\epsilon + O(\epsilon^2)$.

□

Sensitivity of Multiple Eigenvalues and Eigenvectors

- Sensitivity of multiple eigenvalues is more complicated
  - For multiple eigenvalues, left and right eigenvectors can be orthogonal, hence very ill-conditioned
  - In general, multiple or close eigenvalues can be poorly conditioned, especially if matrix is defective
- Condition numbers of eigenvectors are also difficult to analyze
  - If matrix has well-conditioned and well-separated eigenvalues, then eigenvectors are well-conditioned
  - If eigenvalues are ill-conditioned or closely clustered, then eigenvectors may be poorly conditioned

Outline

Algorithms

- QR factorization
  - Classical and modified Gram-Schmidt
  - QR factorization using Householder triangularization and Givens rotation
- Solutions of linear least squares
  - Solution using Householder QR and other QR factorization
  - Alternative solutions: normal equation; SVD
- Conditioning and stability of linear least squares problems
- QR factorization with pivoting

Eigenvalue Problem

- *Eigenvalue problem* of $n \times n$ matrix A is $Ax = \lambda x$
- *Characteristic polynomial* is $\det(A - \lambda I)$
- *Eigenvalue decomposition* of A is $A = X\Lambda X^{-1}$ (does not always exist)
- *Geometric multiplicity* of λ is $\dim(\text{null}(A - \lambda I))$, and *algebraic multiplicity* of λ is its multiplicity as a root of p_A, where algebraic multiplicity $\geq$ geometric multiplicity
- Similarity transformations preserve eigenvalues and their algebraic and geometric multiplicities
- *Schur factorization* $A = QTQ^*$ uses unitary similarity transformations

Eigenvalue Algorithms

- Underlying concepts: power iterations, Rayleigh quotient, inverse iterations, convergence rate
- *Schur factorization* is typically done in two steps
  - Reduction to Hessenberg form for non-Hermitian matrices or reduction to tridiagonal form for hermitian matrices by unitary similarity transformation
  - Finding eigenvalues of Hessenberg or **tridiagonal** form
- Finding eigenvalue of tridiagonal forms
  - QR algorithm with shifts, and their interpretations as (inverse) simultaneous iterations
- Generalized eigenvalue problems for symmstric matrices